

Jiangxi University of Finance and Economics

Relational Schema & Zachman Framework Report

Luckin Coffee Inc.

A Technology-Driven New Retail Coffee Enterprise

IK2015: Enterprise Architecture and Database Programming

Seminar 5

Group 4

Zhe-Kai Yang (Krust), Shi-Hang Chen (Sean), Wei-Yu Cai (Nosta),

Huan-Wen Chen (Evan), Yuan-Yong Liao (Foreve)

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1 Report Basic Information

Table 1: Basic information of the relational schema and Zachman framework report.

Item	Information
Selected enterprise / business	Luckin Coffee Inc.; mobile-first coffee retail, pickup, delivery, partnership-store operation, and packaged coffee business.
Course and seminar	IK2015 Enterprise Architecture and Database Programming; Seminar 5.
Group number	Group 4.
Group members	Zhe-Kai Yang (Krust), Shi-Hang Chen (Sean), Wei-Yu Cai (Nosta), Huan-Wen Chen (Evan), Yuan-Yong Liao (Foreve).
Main deliverable	Zachman Framework Row 1 (Scope) description for Luckin Coffee and a relational database schema mapping the conceptual model from Seminar 4, complete with primary/foreign keys and relationships.

This report fulfills the two main requirements of Seminar 5: describing the first row of the Zachman Framework (Scope/Contextual Planner) for our case study, and transforming the conceptual model created in Seminar 4 into a formal relational database schema.

2 Zachman Framework: Row 1 (Scope/Contextual Planner)

The Zachman Framework is a schema for enterprise architecture that structures architectural deliverables along six perspectives (rows) and six interrogatives (columns). The first row, **Scope (Contextual)**, represents the perspective of the ****Planner****. It establishes the business context, boundaries, and high-level inventory of elements that are relevant to the enterprise's strategy and operations.

Below is the detailed description of Row 1 for Luckin Coffee Inc. across the six interrogative columns.

Table 2: Zachman Framework Row 1 (Scope / Planner) for Luckin Coffee Inc.

	DATA (What?)	FUNCTION (How?)	NETWORK (Where?)	PEOPLE (Who?)	TIME (When?)	MOTIVATION (Why?)
SCOPE (Contextual) Planner	- Customers & Members - Digital Channels - Orders & Items - Products & Recipes - Stores & Staff - Raw Materials & Suppliers - Performance KPIs	- Customer Acquisition - Mobile Order & Payment - Store-level Fulfillment - R&D & Recipe Mgmt - Supply Chain - Logistics Handover	- Corporate HQ - Logistics Hubs - Supplier Origins - Retail Grid - Digital Cloud Network - Consumption Zones	- Digital-native Customers - Baristas & Store Managers - Corporate Planners (Partners) - Franchisees - Material Suppliers - Third-party Riders	- Real-Time Fulfillment - Daily Store Hours - Diurnal Peak Hours - Weekly Replenishment - Fast R&D Cycles	- Mobile-First / New Retail - Affordable Premium - Rapid Market Penetration - Strict Standardization - CLV Maximization

3 Relational Database Schema Design

To implement the conceptual model from Seminar 4, we define a relational database schema. The schema maps entities to primary tables, handles 1:1 and 1:N associations with foreign keys, and resolves M:N relationships through bridge tables.

3.1 Table Definitions and Attributes

Underlined columns represent the ****Primary Key (PK)**** of the table. Columns appended with '(FK)' refer to a primary key in another table.

Table 3: Relational Database Schema Table Definitions.

Table Name	Attributes (Primary Keys underlined, Foreign Keys indicated)
Customers	<u>CustomerId</u> , Segment, City, AgeBand
MembershipAccounts	<u>MemberId</u> , CustomerId (FK, Unique), PointsBalance, CouponValue, MemberTier, ActivityStatus
DigitalChannels	<u>ChannelId</u> , ChannelType, AccessDevice, AppVersion
PromotionCampaigns	<u>CampaignId</u> , CampaignType, TargetSegment, ValidPeriod, SubsidyAmount
MemberCampaigns	<u>MemberId</u> (FK), <u>CampaignId</u> (FK), ReceivedDate, UsedStatus
DigitalOrders	<u>OrderId</u> , MemberId (FK), ChannelId (FK), OrderChannel, PaymentMethod, QueueStatus, OrderAmount, DeliveryFee, OrderDate, OrderTime
OrderCampaigns	<u>OrderId</u> (FK), <u>CampaignId</u> (FK), AppliedAmount
OrderItems	<u>OrderId</u> (FK), <u>ItemNr</u> , ProductId (FK), Quantity, Customization, Discount, ItemPrice
Products	<u>ProductId</u> , ProductName, Category, Season, GrossMargin, StandardCost
RecipeStandards	<u>RecipeId</u> , ProductId (FK), RecipeVersion, SopStatus, StandardCost, DefectRate
RecipeMaterials	<u>RecipeId</u> (FK), <u>MaterialId</u> (FK), Dosage
RawMaterials	<u>MaterialId</u> , SupplierId (FK), MaterialName, MaterialType, Origin, BatchStatus, InventoryValue
Suppliers	<u>SupplierId</u> , SupplierName, SupplierType, LeadTime
Stores	<u>StoreId</u> , StoreName, OwnershipType, CityTier, LocationType, Revenue, LaborCost
StoreStaff	<u>StaffId</u> , StoreId (FK), StaffName, StaffRole, TrainingStatus, WorkingHours, LaborCost, Productivity
StoreMaterials	<u>StoreId</u> (FK), <u>MaterialId</u> (FK), StockLevel, StockoutRate
FulfilmentTasks	<u>TaskId</u> , OrderId (FK), StoreId (FK), PartnerId (FK, Nullable), TaskType, PrepTime, WaitingTime, CompletionRate, Status
DeliveryPartners	<u>PartnerId</u> , PartnerName, PartnerType, ServiceArea, DeliveryFee, ComplaintRate

3.2 Conceptual to Relational Mapping Logic

The transition from the conceptual model in Seminar 4 to the relational database schema relies on standard mapping rules:

- **Entity Mapping:** Business objects (e.g., Customers, Products, Stores) map to separate

relation tables, with a unique identifier acting as the Primary Key (PK).

- **1:1 Relationships:** The customer-membership association (D2) is mapped by adding a unique foreign key 'CustomerId (FK)' to the 'MembershipAccounts' table. This enforces that a member is tied to exactly one customer.
- **1:N Relationships:** Relationships such as 'DigitalOrders – OrderItem' (F1) and 'Store – StoreStaff' (F4) are represented by placing the parent's primary key ('OrderId' or 'StoreId') as a Foreign Key (FK) in the child table. For order items, the primary key is composite '(OrderId, ItemNr)' to support multiple lines per order.
- **M:N Relationships:** Relational databases cannot store multi-valued lists directly. Therefore, many-to-many associations are resolved by creating **bridge tables** that contain composite primary keys formed by the primary keys of both participating tables:
 - 'RecipeMaterials' connects 'RecipeStandards' and 'RawMaterials' (S3) and records the specific 'Dosage' of each ingredient in a drink.
 - 'StoreMaterials' tracks the current inventory level ('StockLevel') of each raw material within a specific store (S5).
 - 'MemberCampaigns' records the coupons distributed to members (D4).
 - 'OrderCampaigns' links promotion campaigns to digital orders to trace which discounts were applied to a specific purchase (D5).

4 Relational Database Schema Diagram

Luckin Coffee Inc. Relational Database Schema Diagram

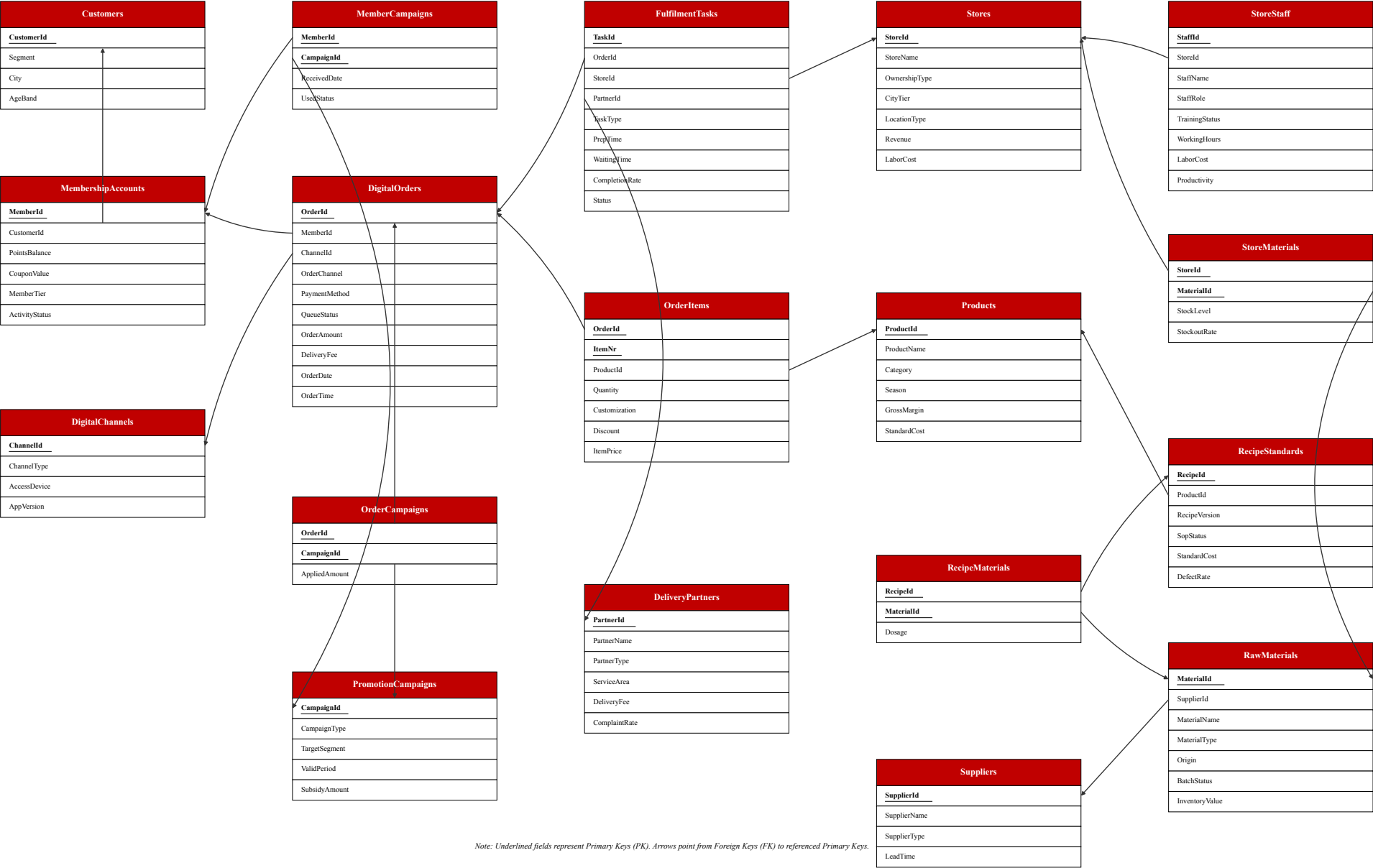


Figure 1: Relational database schema diagram for Luckin Coffee. The tables are shown as grid structures where the column headers are colored red. Underlined column names indicate Primary Keys (PK). Solid lines with arrows point from the referencing Foreign Key (FK) to the referenced Primary Key (PK).

5 Business Scenarios Supported by the Relational Schema

To demonstrate the completeness of the relational database design, we trace the SQL joining logic for three core retail scenarios.

5.1 Scenario 1: Mobile Order and Pickup Fulfilment

When a customer logs in, uses a coupon, submits a coffee order, and picks it up at a store:

- **SQL Joins:** ‘Customers’ join ‘MembershipAccounts’ on ‘CustomerId’ → join ‘MemberCampaigns’ and ‘PromotionCampaigns’ on ‘MemberId’ / ‘CampaignId’ → join ‘DigitalOrders’ on ‘MemberId’ → join ‘OrderItems’ and ‘Products’ on ‘OrderId’ / ‘ProductId’ → join ‘FulfilmentTasks’ on ‘OrderId’ → join ‘Stores’ on ‘StoreId’.
- **Data Flow:** Traces customer acquisition, coupon application, beverage customization, barista preparation time, and store collection.

5.2 Scenario 2: R&D Product Launch and Margin Auditing

When R&D releases a new beverage series and corporate analysts evaluate actual gross margins vs. standard ingredient costs:

- **SQL Joins:** ‘Products’ join ‘RecipeStandards’ on ‘ProductId’ → join ‘RecipeMaterials’ on ‘RecipeId’ → join ‘RawMaterials’ on ‘MaterialId’ → join ‘Suppliers’ on ‘SupplierId’. Additionally, join ‘Products’ to ‘OrderItems’ on ‘ProductId’ to pull actual transaction discounts.
- **Data Flow:** Allows the system to compare the standard cost (sum of ingredient dosages × raw material purchase cost) against the actual net revenue (item price minus custom discounts) to audit gross margins.

5.3 Scenario 3: Store Inventory Replenishment

When a store’s stock of coffee beans falls below the safety threshold, triggering an automated replenishment order:

- **SQL Joins:** ‘Stores’ join ‘StoreMaterials’ on ‘StoreId’ → join ‘RawMaterials’ on ‘MaterialId’ → join ‘Suppliers’ on ‘SupplierId’.
- **Data Flow:** The system queries ‘StoreMaterials’ where ‘StockLevel’ < safety threshold, aggregates required quantities, groups them by ‘SupplierId’, and submits purchase orders to suppliers.

6 Conclusion

This report successfully mapped the strategic and conceptual architecture of Luckin Coffee to a relational database implementation.

First, describing Row 1 of the Zachman Framework established the strategic context by outlining what data is critical, how processes operate, where nodes are located, who the stakeholders

are, when events occur, and why specific strategies are chosen.

Second, the relational database design translated the conceptual objects from Seminar 4 into 18 primary and bridge tables. By defining primary keys, foreign keys, and structural mapping rules, the database schema supports critical business operations, including mobile ordering, recipe control, inventory management, and promotional auditing. The schema diagram visually establishes the structural integrity of the database, providing a robust blueprint for enterprise architecture and database programming.

References

- [1] Zachman, J. A. (1987). *A Framework for Information Systems Architecture*. IBM Systems Journal, 26(3), 276–292.
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- [4] Elmasri, R., & Navathe, S. B. (2015). *Fundamentals of Database Systems* (7th ed.). Pearson.